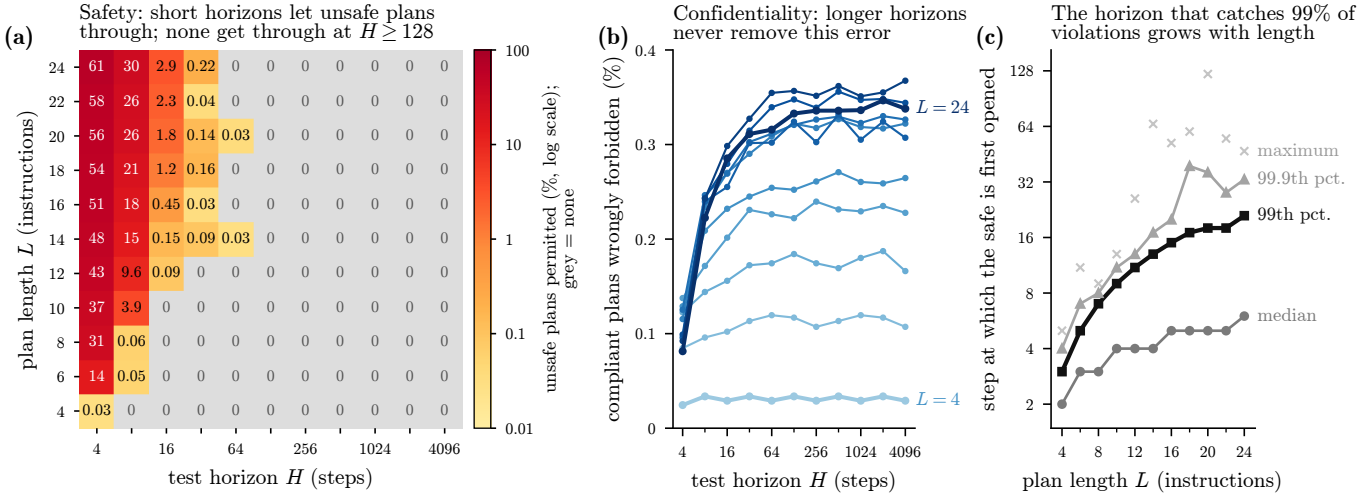
(a)
Safety: short horizons let unsafe plans through; none get through at $H \geq 128$
plan length L (instructions)
test horizon H (steps)
unsafe plans permitted (%, log scale); grey = none
100
10
1
0.1
0.01
24 61 30 2.9 0.22 0 0 0 0 0 0 0
22 58 26 2.3 0.04 0 0 0 0 0 0 0
20 56 26 1.8 0.14 0.03 0 0 0 0 0 0
18 54 21 1.2 0.16 0 0 0 0 0 0 0
16 51 18 0.45 0.03 0 0 0 0 0 0 0
14 48 15 0.15 0.09 0.03 0 0 0 0 0 0
12 43 9.6 0.09 0 0 0 0 0 0 0 0
10 37 3.9 0 0 0 0 0 0 0 0 0
8 31 0.06 0 0 0 0 0 0 0 0 0
6 14 0.05 0 0 0 0 0 0 0 0 0
4 0.03 0 0 0 0 0 0 0 0 0 0
4 16 64 256 1024 4096
(b)
Confidentiality: longer horizons never remove this error
compliant plans wrongly forbidden (%)
0.4
0.3
0.2
0.1
0
$L = 24$
$L = 4$
4 16 64 256 1024 4096
test horizon H (steps)
(c)
The horizon that catches 99% of violations grows with length
step at which the safe is first opened
128
64
32
16
8
4
2
maximum
99.9th pct.
99th pct.
median
4 8 12 16 20 24
plan length L (instructions)